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Report: SDM using Random Forest

4.1 Why did you choose this species?

Ans: I chose the European hedgehog because it is a widely distributed and ecologically important mammal in Europe, currently facing population declines due to habitat loss and environmental changes. Modeling its distribution can help inform conservation strategies.

4.2 Why were the environmental variables chosen?

Ans: Do they have any influence in their distribution?
The selected WorldClim variables,annual mean temperature (bio1), maximum temperature of warmest month (bio5), minimum temperature of coldest month (bio6), and annual precipitation (bio12), are key climatic factors that influence hedgehog survival, hibernation, and food availability. These variables are known to affect the distribution of many terrestrial mammals, including hedgehogs.

4.3 How does the model perform? AUC for the training and test datasets?

Ans: The model performs well, with a training AUC of 0.895 and a test AUC of 0.749. This indicates good discrimination between suitable and unsuitable habitats, though the lower test AUC compared to training AUC suggests some overfitting.

4.4 Explain the variable importance for your species ecologically.

Ans: According to the variable importance results, annual precipitation (bio12) and minimum temperature of the coldest month (bio6) are the most important predictors. This makes ecological sense, as hedgehogs require moist environments for foraging and are sensitive to cold temperatures during hibernation. Temperature extremes and precipitation patterns directly affect their survival and distribution.

4.5 Explain the current habitat suitability map.

Ans: The habitat suitability map shows areas in Germany predicted to be most suitable for the European hedgehog, based on the selected climate variables. High suitability areas correspond to regions with moderate temperatures and adequate precipitation, matching known hedgehog habitats. The map can help identify priority areas for conservation and further study.